CIRC HQ STUDENT HANDOUT | GRADE 5

Moving Picture Machine

Goal: Create a sequence device that uses small image changes to produce the illusion of motion.

35 minutes | 8 steps | Engineering and Design | Playbook A meeting 19, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Run the animation and identify one frame changed to make the motion smoother.

You will use

- Index cards
- Paper strips
- Cardstock circles
- Pencils
- Markers
- Brass fasteners
- Binder clips

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Choose flip, wheel, or strip motion.
- Place the frame paper in order.

2 Step 2: Fastener safety — 3 min

- Keep the device at table level.
- Use fasteners only through marked holes.
- Ask the teacher about a sharp fastener.

3 Step 3: Plan the motion — 4 min

- Draw a clear start pose.
- Draw a clear finish pose.
- Mark the matching dot.

4 Step 4: Draw the frames — 9 min

- Change one small part per frame.
- Keep each drawing on the registration mark.
- Number the frames.

5 Step 5: Assemble the motion — 7 min

- Stack or attach the frames in order.
- Secure the approved fastener.
- Move the device slowly.

6 Step 6: Smooth the motion — 4 min

- Find the largest visual jump.
- Revise one frame.
- Test at two speeds.

7 Step 7: Sort the pieces — 2 min

- Return unused fasteners.
- Clip your frames together.

8 Step 8: Exit check — 3 min

- Show the finished motion once.
- Name the frame you changed.

What to turn in

A running animation with one smoothed frame.

No computer? Another way to do it

A paper flip book of the same frames.

Finished early? Next challenge

Smooth Loop: Add transition frames so the final image flows naturally back to the first.

Stay safe

Keep spinning devices at table level and handle fasteners only as demonstrated.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.